Oral cavity & tongue

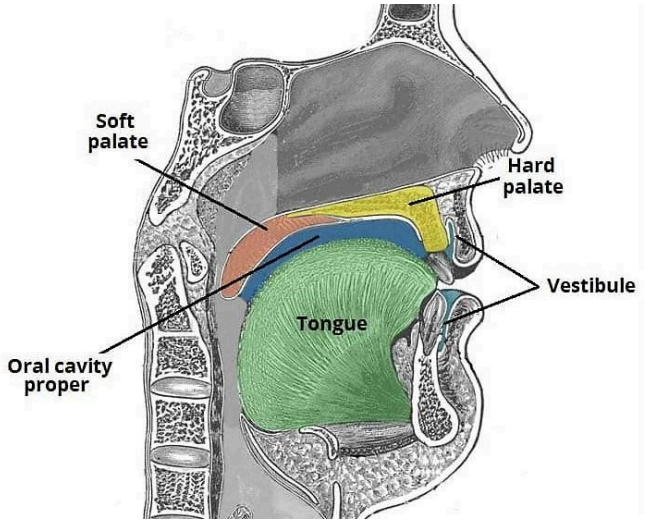
Boundaries and cancer subsites of the oral cavity, lips, floor of mouth, retromolar trigone, palate and tongue with its muscles, nerves and lymphatics.

1. Boundaries and subsites
2. Lips
3. Floor of mouth, buccal mucosa, retromolar trigone, palate
4. Tongue: parts and papillae
5. Muscles of the tongue
6. Nerves, taste and lymphatics of the tongue
7. Soft palate
8. In theatre
9. On imaging

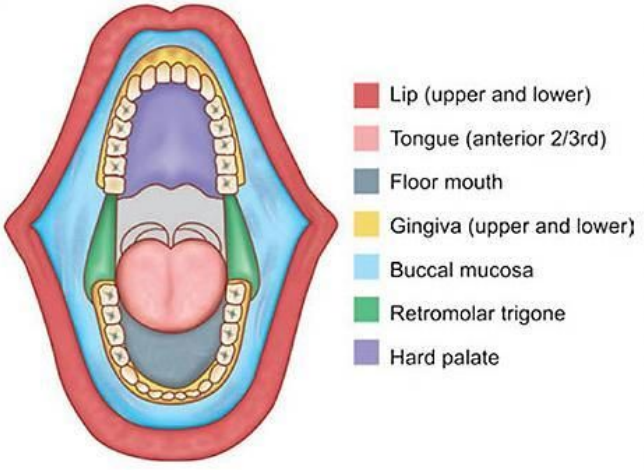
01 Boundaries and subsites

- From the **lips** to the junction of the **hard and soft palate** above and the **circumvallate papillae** below. Divided into the **vestibule** and the **oral cavity proper**.
- **Cancer subsites:** lip, buccal mucosa, upper and lower alveolus and gingiva, retromolar trigone, floor of mouth, oral tongue (anterior two-thirds), hard palate.
- The soft palate and tongue base belong to the **oropharynx**.

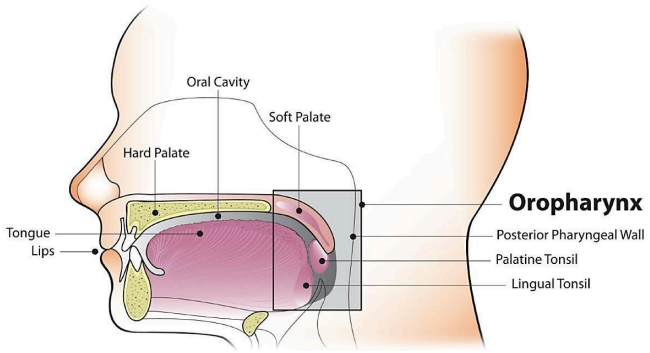
CLINICAL PEARL Tumours of the oral cavity and oropharynx are staged differently, so place the tumour precisely relative to the circumvallate papillae and hard–soft palate junction.



Oral cavity: vestibule and oral cavity proper
ENT Steps Anatomy, p. 164

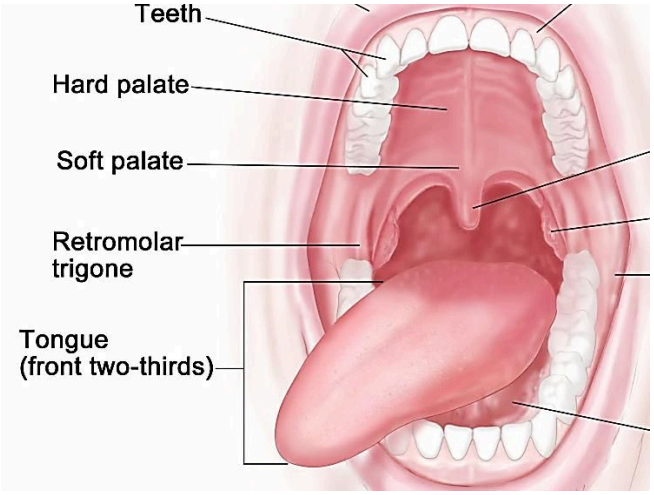


Subsites of the oral cavity
ENT Steps Anatomy, p. 165



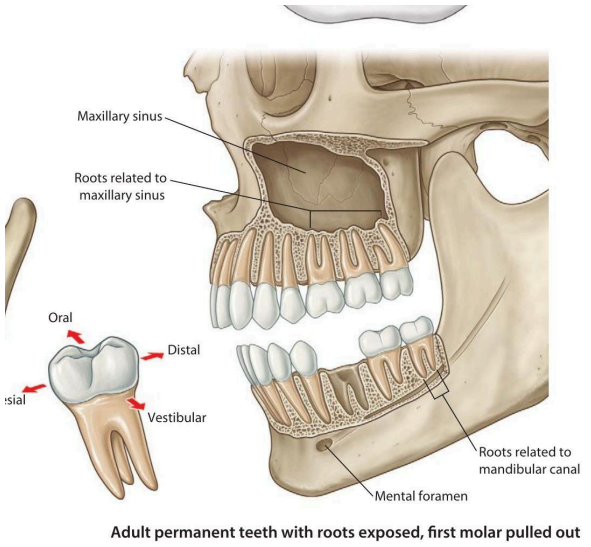
Boundary between the oral cavity and oropharynx

ENT Steps Anatomy, p. 165



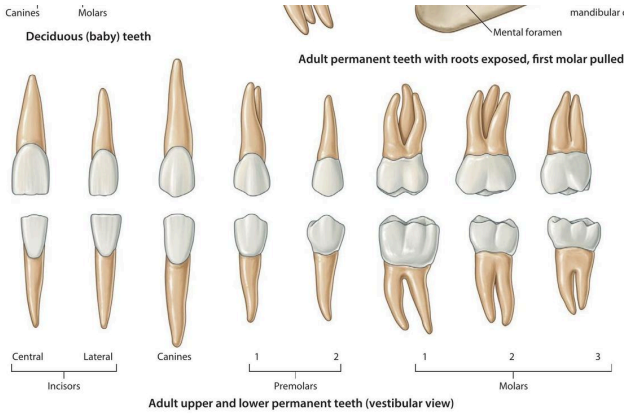
Hard palate, soft palate, retromolar trigone and oral tongue

ENT Steps Anatomy, p. 172



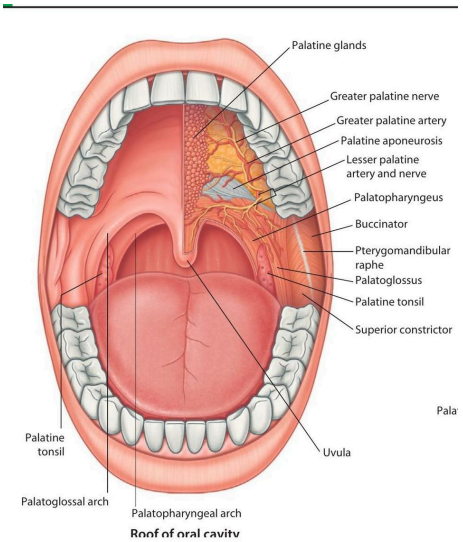
Adult permanent teeth with roots exposed

Gray's Atlas of Anatomy, 2nd ed., p. 573



Adult upper and lower permanent teeth

Gray's Atlas of Anatomy, 2nd ed., p. 573



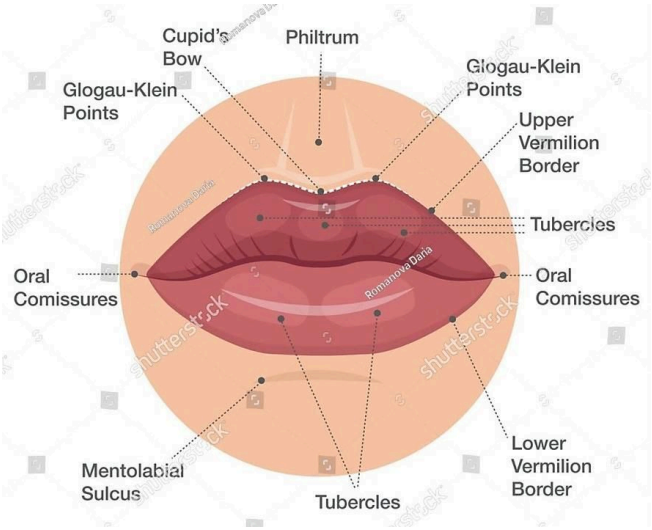
Roof of the oral cavity

Gray's Atlas of Anatomy, 2nd ed., p. 582

02 Lips

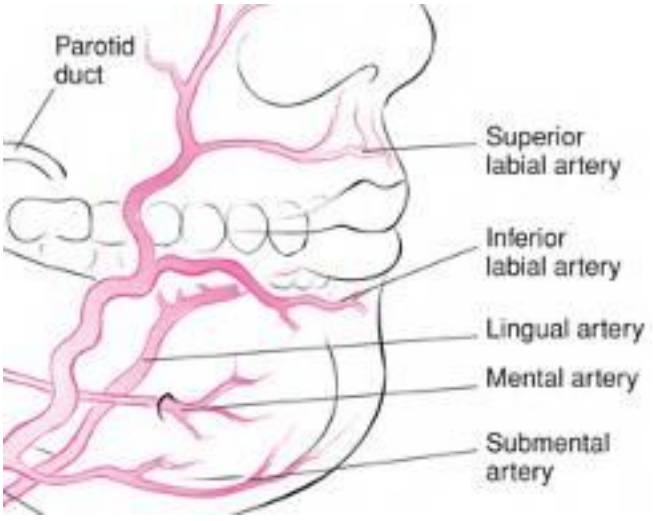
- **Vermilion:** non-keratinised, transparent epithelium without hair or glands, so blood colour shows through. The dry vermilion meets skin at the vermilion border (Cupid's bow above).
- Mucosa is only about 2 mm from the orbicularis oris.
- **Blood:** superior and inferior labial arteries (facial).
- **Lymph:** central lower lip → **submental (IA)**; lateral lower lip and upper lip → **submandibular (IB)**; upper lip also to preauricular and infraparotid nodes.

CLINICAL PEARL Lower lip cancer near the midline can drain to both sides.



Lip landmarks: vermilion, philtrum, Cupid's bow, commissures

ENT Steps Anatomy, p. 166



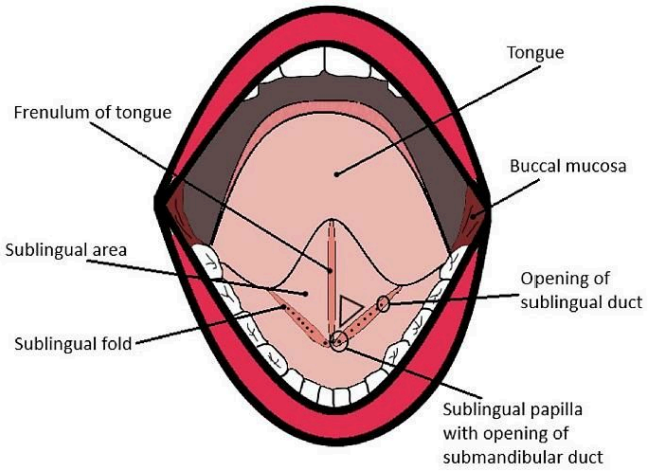
Arterial supply of the lips

ENT Steps Anatomy, p. 168

03 Floor of mouth, buccal mucosa, retromolar trigone, palate

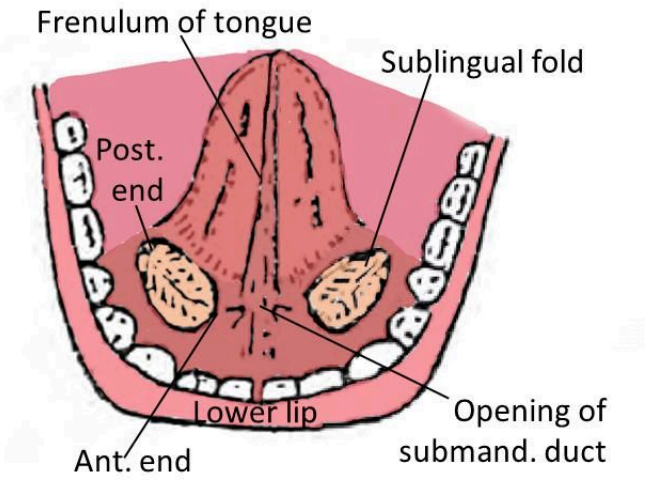
- **Floor of mouth:** crescent between the lower gum and the tongue's undersurface; **frenulum**, **sublingual papillae** (submandibular duct openings) and sublingual folds. Anterior floor lymphatics **cross the midline** to IA/IB; posterior floor to levels II–III.
- **Buccal mucosa:** lines the cheeks between the gingivobuccal sulci, back to the **pterygomandibular raphe**; the parotid duct opens opposite the **upper second molar**.
- **Pterygomandibular raphe:** from the pterygoid hamulus to the mylohyoid line; origin of **buccinator** in front and **superior constrictor** behind. Landmark for the inferior alveolar nerve block.
- **Retromolar trigone:** mucosa over the front of the ramus behind the last molar; tumours here reach bone, the pterygoid muscles and the lingual and inferior alveolar nerves early.
- **Hard palate:** many minor salivary glands; anterior part drains to submandibular nodes, posterior to retropharyngeal and upper deep cervical nodes.

CLINICAL PEARL The hard palate is the commonest site for minor salivary gland tumours.



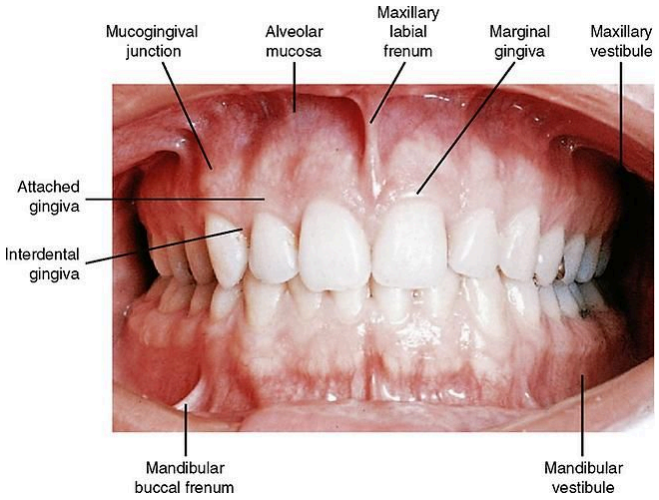
Floor of the mouth and sublingual papillae

ENT Steps Anatomy, p. 169



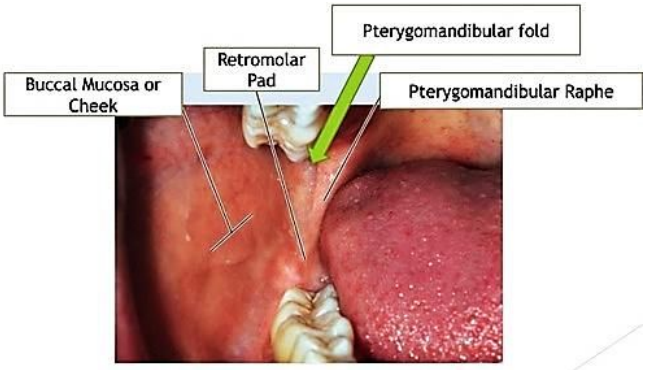
Floor of the mouth: sublingual fold and duct openings

Dr. Sameh Doss handout, p. 121



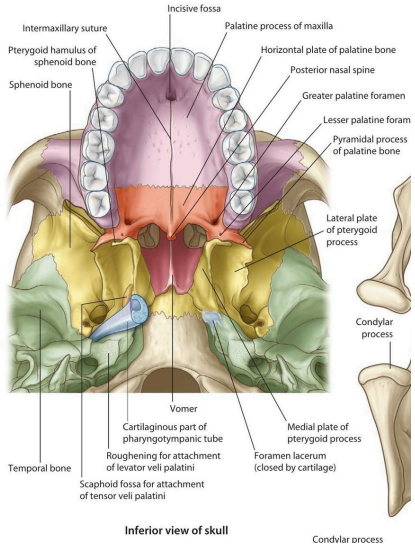
Gingiva and alveolar mucosa

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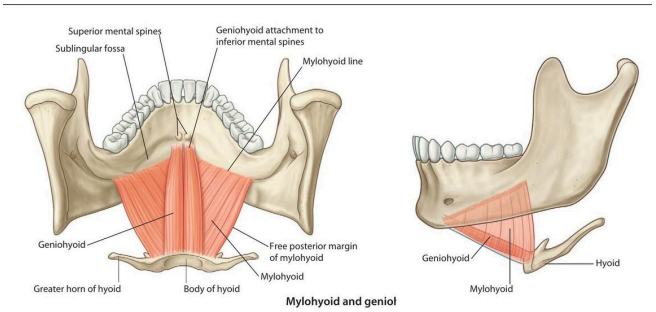
Retromolar trigone and pterygomandibular raphe

ENT Steps Anatomy, p. 171



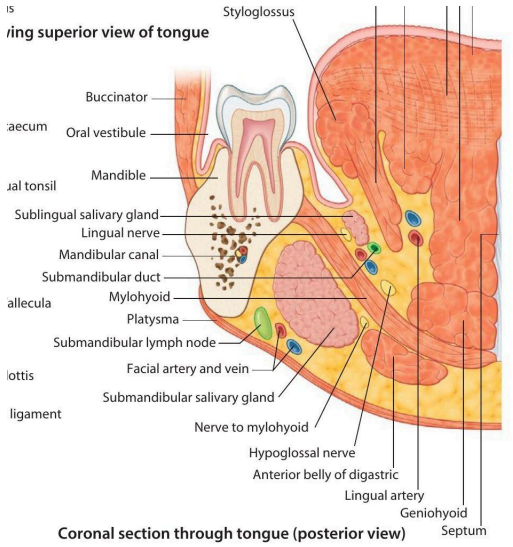
Inferior view of the skull: hard palate and alveolar arch

Gray's Atlas of Anatomy, 2nd ed., p. 572



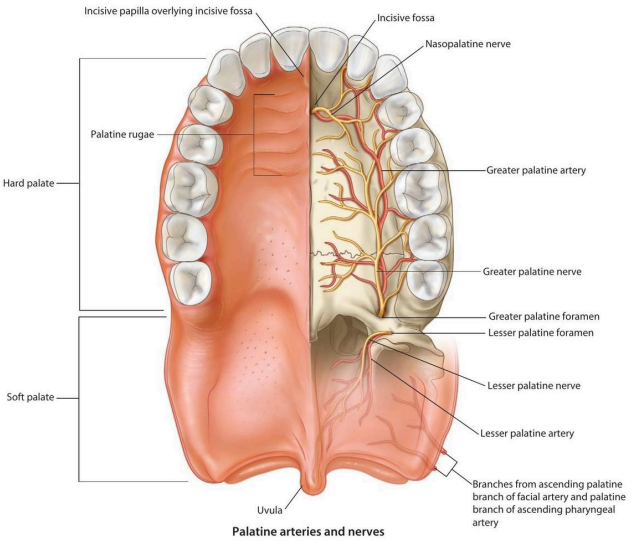
Mylohyoid and geniohyoid

Gray's Atlas of Anatomy, 2nd ed., p. 578



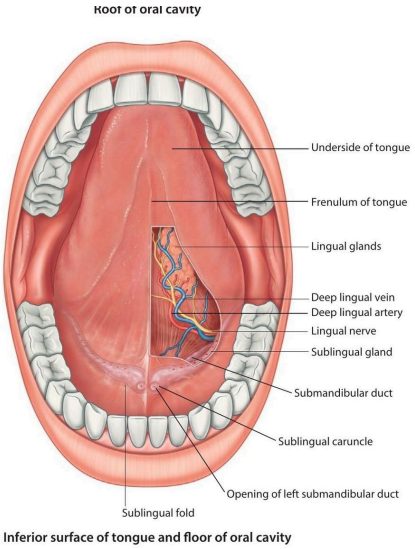
Coronal section through the tongue and floor of mouth

Gray's Atlas of Anatomy, 2nd ed., p. 580



Palatine arteries and nerves

Gray's Atlas of Anatomy, 2nd ed., p. 581

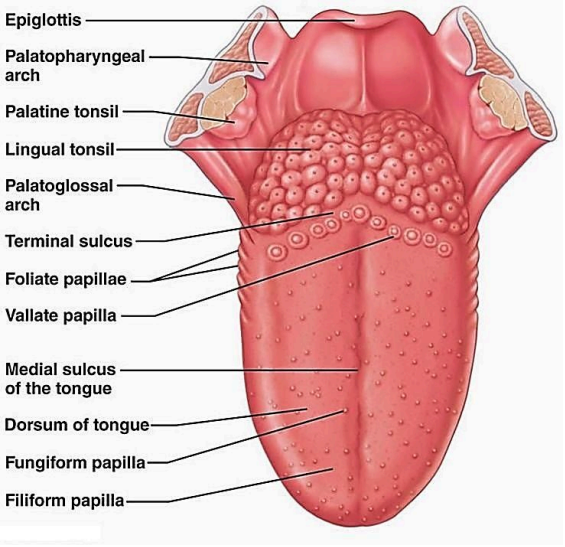


Inferior surface of the tongue and floor of the oral cavity

Gray's Atlas of Anatomy, 2nd ed., p. 582

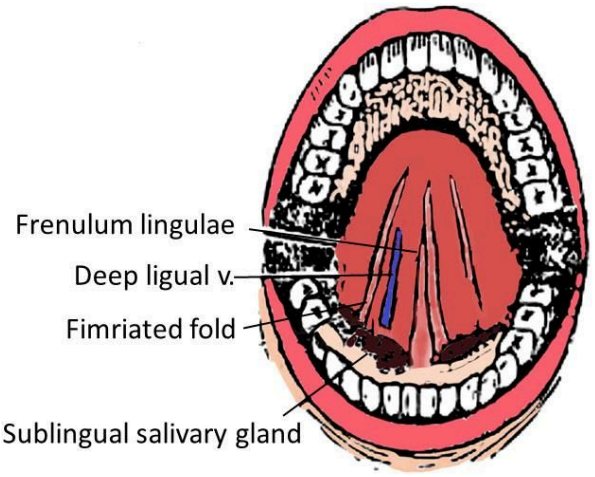
04 Tongue: parts and papillae

- **Oral tongue** (anterior two-thirds: tip, borders, dorsum, ventral surface) in front of the **sulcus terminalis**; **base** (posterior third) faces the oropharynx with the **lingual tonsil**.
- **Foramen caecum** at the apex of the sulcus terminalis: origin of the thyroglossal duct.
- **Papillae:** circumvallate (8–12, in front of the sulcus, most taste buds), foliate (lateral borders), fungiform (tip and borders), filiform (most numerous, no taste buds).
- Undersurface: frenulum, deep lingual veins and fimbriated folds.



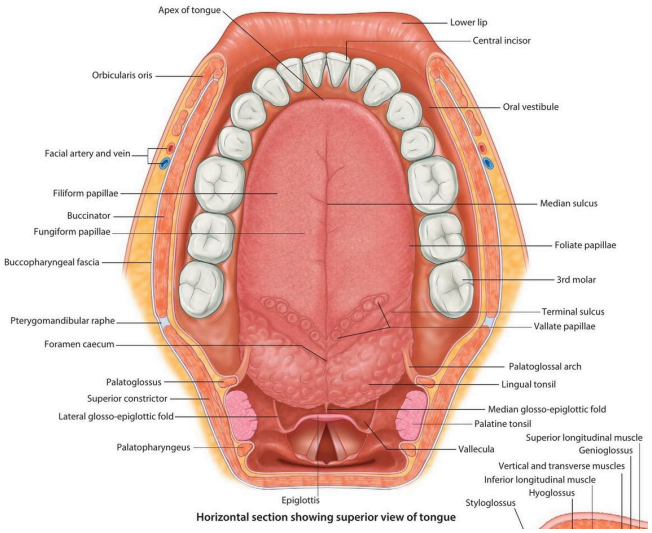
Dorsum of the tongue: papillae, sulcus terminalis, lingual tonsil

ENT Steps Anatomy, p. 173



Undersurface of the tongue and floor of the mouth

Dr. Sameh Doss handout, p. 177



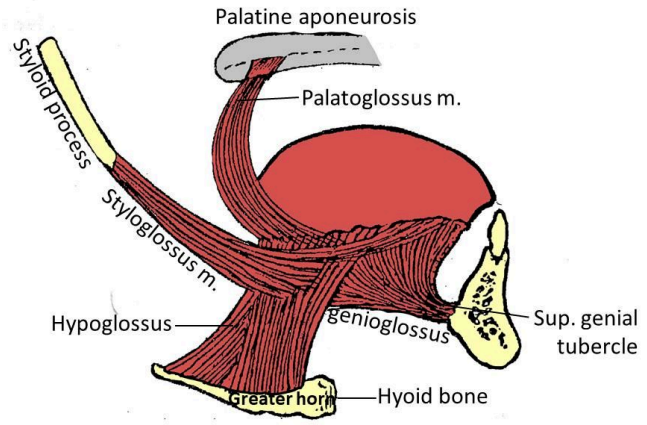
Horizontal section: superior view of the tongue and papillae

Gray's Atlas of Anatomy, 2nd ed., p. 580

05 Muscles of the tongue

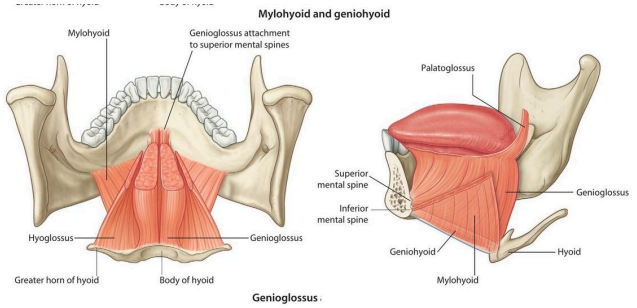
- **Intrinsic:** longitudinal, transverse and vertical fibres change the shape of the tongue.
- **Extrinsic: genioglossus** (protrudes; the safety muscle that keeps the tongue off the airway), **hyoglossus** (depresses), **styloglossus** (retracts and elevates), **palatoglossus** (elevates the back).
- All supplied by the **hypoglossal nerve (XII)** except **palatoglossus (vagus via the pharyngeal plexus)**.

> **CLINICAL PEARL** Unilateral XII palsy: the tongue deviates toward the weak side on protrusion (the healthy genioglossus pushes it over).



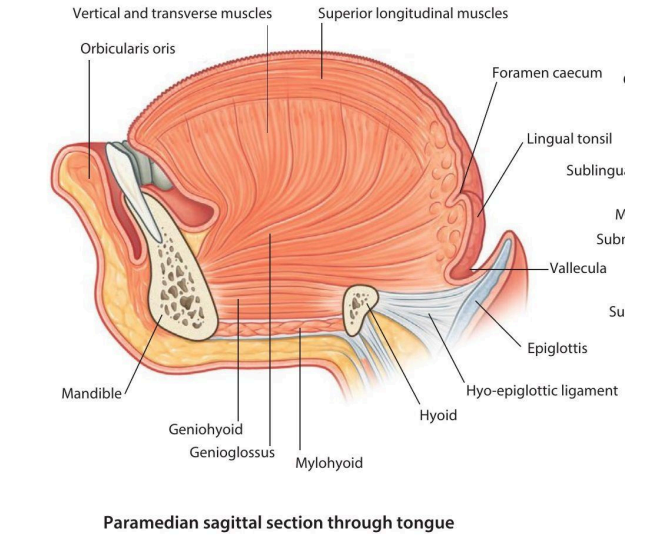
Extrinsic muscles of the tongue

Dr. Sameh Doss handout, p. 179



Genioglossus and hyoglossus

Gray's Atlas of Anatomy, 2nd ed., p. 578



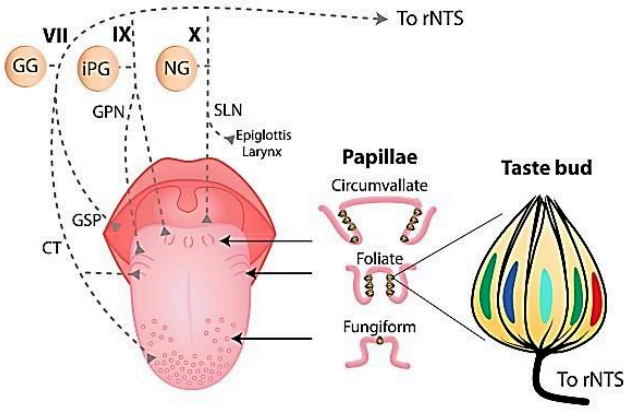
Paramedian sagittal section through the tongue

Gray's Atlas of Anatomy, 2nd ed., p. 580

06 Nerves, taste and lymphatics of the tongue

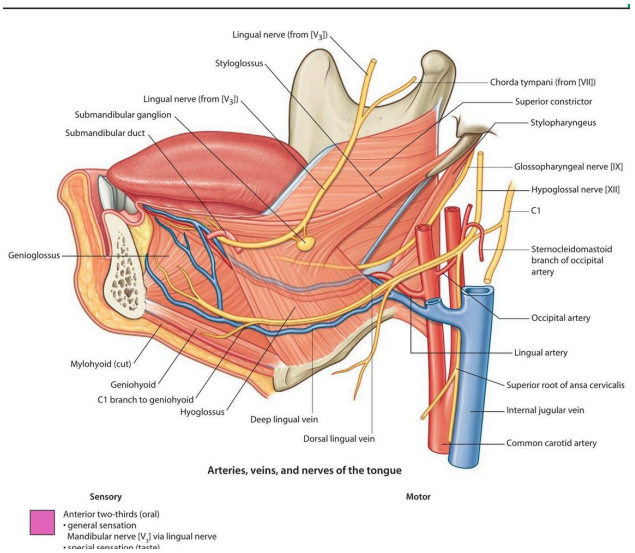
- **Anterior two-thirds:** general sensation by the **lingual nerve (V3)**; taste by the **chorda tympani (VII)**.
- **Posterior third:** general sensation and taste by **glossopharyngeal (IX)**.
- **Vallecula and epiglottis:** internal laryngeal nerve (X).
- **Lymph:** tip → **submental**; lateral borders → **submandibular** and deep cervical (can skip to **level III–IV**, the jugulo-omohyoid node); central part and base → **both sides** of the neck.

> **CLINICAL PEARL** Lateral tongue cancer can metastasise straight to level III–IV, so selective neck dissection for oral tongue includes levels I–IV.



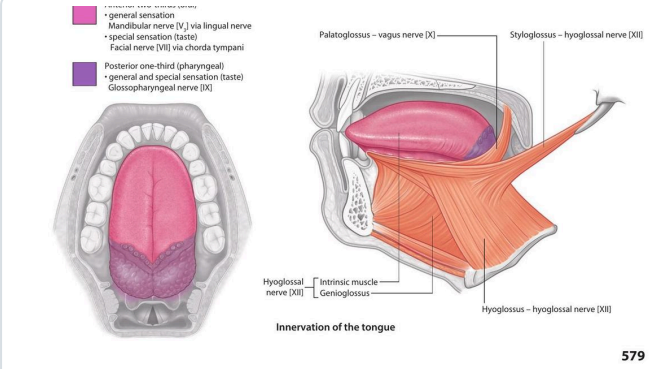
Nerve supply and taste pathways of the tongue

ENT Steps Anatomy, p. 174



Arteries, veins and nerves of the tongue

Gray's Atlas of Anatomy, 2nd ed., p. 579



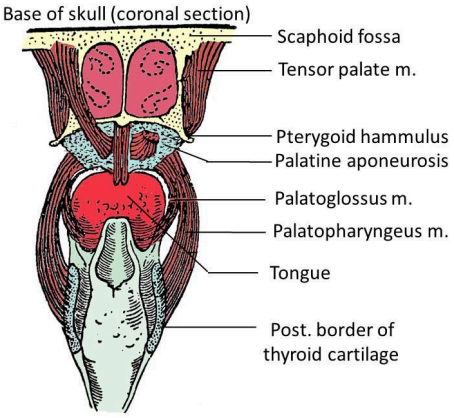
Innervation of the tongue: sensory and motor

Gray's Atlas of Anatomy, 2nd ed., p. 579

07 Soft palate

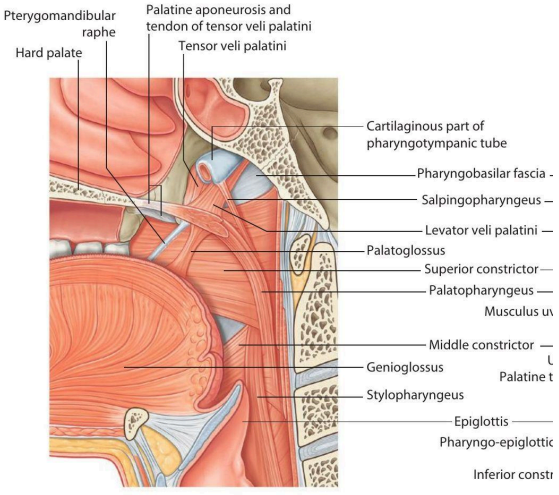
- Palatine aponeurosis plus five muscles: **tensor veli palatini (V3)**, **levator veli palatini**, palatoglossus, palatopharyngeus and musculus uvulae (all **vagus via the pharyngeal plexus**).
- **Tensor** hooks round the pterygoid hamulus and opens the ET; **levator** lifts the palate to close the nasopharynx.

> **CLINICAL PEARL** Cleft palate disrupts the tensor's pull on the ET, which is why most children with cleft palate have otitis media with effusion.



Muscles of the soft palate

Dr. Sameh Doss handout, p. 181



Muscles of the soft palate (sagittal section)

Muscles of the soft palate (sagittal section)

Gray's Atlas of Anatomy, 2nd ed., p. 581

08 Surgical anatomy: In theatre

- **Access:** transoral for small tumours; lip split with mandibulotomy (paramedian, in front of the mental foramen) or a visor flap for larger ones.
- **Margins:** aim for about 1 cm clinically (5 mm histologically); depth of invasion drives elective neck dissection (over about 3–4 mm).
- **Mandible:** marginal mandibulectomy when tumour abuts but does not invade bone; segmental when the medulla is invaded.
- Protect the **lingual nerve** and **Wharton's duct** in the floor of the mouth; the hypoglossal nerve lies deeper on hyoglossus.

09 Radiological anatomy: On imaging

- **MRI** shows tongue and floor of mouth tumour depth and spread across the midline; **CT** and **OPG** show mandibular invasion.
- **Ranula:** a plunging ranula has a tail through or behind the mylohyoid into the submandibular space.

T Tables

Tongue: nerves

Part	General sensation	Taste	Motor
Anterior two-thirds	Lingual nerve (V3)	Chorda tympani (VII)	XII (all except palatoglossus)
Posterior third	Glossopharyngeal (IX)	Glossopharyngeal (IX)	—
Vallecula, epiglottis	Internal laryngeal (X)	Internal laryngeal (X)	—
Palatoglossus	—	—	Pharyngeal plexus (X)

Cleft lip and palate: anatomy

Type	Structures involved	Cause
Cleft lip	Lip ± alveolus, in front of incisive foramen	Maxillary + medial nasal prominence failure
Cleft palate (secondary)	Behind incisive foramen	Palatal shelves fail to fuse

Type	Structures involved	Cause
Complete cleft lip and palate	Both, through the alveolus	Both of the above
Submucous cleft	Bifid uvula, hard palate notch, zona pellucida	Muscle not united under intact mucosa

Muscles of the soft palate

Muscle	Origin	Insertion	Nerve	Action
Tensor veli palatini	Scaphoid fossa, spine of sphenoid, cartilaginous Eustachian tube	Palatine aponeurosis (tendon hooks round the hamulus)	V3 (via nerve to medial pterygoid)	Tenses palate; opens the Eustachian tube
Levator veli palatini	Petrous temporal bone in front of carotid canal; tube cartilage	Upper surface of palatine aponeurosis	Vagus (pharyngeal plexus)	Only elevator of the soft palate above neutral
Palatopharyngeus	Upper surface of palatine aponeurosis	Pharyngeal wall	Vagus (pharyngeal plexus)	Depresses palate; narrows isthmus; elevates pharynx
Palatoglossus	Under surface of palatine aponeurosis	Side of tongue	Vagus (pharyngeal plexus)	Depresses palate; elevates back of tongue
Musculus uvulae	Posterior nasal spine	Mucosa of uvula	Vagus (pharyngeal plexus)	Elevates and retracts uvula

Gray's Atlas. Tensor is the only palatal muscle not supplied by the pharyngeal plexus.

Muscles of the tongue

Muscle	Origin	Insertion	Nerve	Action
Superior longitudinal (intrinsic)	Submucosa at back of tongue, median septum	Margins of tongue	XII	Shortens tongue; curls apex and sides up
Inferior longitudinal (intrinsic)	Root of tongue, hyoid	Apex	XII	Shortens tongue; turns apex down
Transverse (intrinsic)	Median septum	Lateral margins	XII	Narrows and lengthens tongue
Vertical (intrinsic)	Dorsal submucosa	Ventral connective tissue	XII	Flattens and widens tongue
Genioglossus	Superior mental spines	Whole length of tongue; body of hyoid	XII	Protrudes tongue; depresses centre
Hyoglossus	Greater horn and body of hyoid	Side of tongue	XII	Depresses tongue
Styloglossus	Styloid process	Side of tongue	XII	Retracts and elevates tongue
Palatoglossus	Palatine aponeurosis	Side of tongue	Vagus (pharyngeal plexus)	Elevates back of tongue; narrows isthmus

Gray's Atlas. All tongue muscles are XII except palatoglossus (X).

Suprahyoid and infrahyoid muscles

Muscle	Origin	Insertion	Nerve	Action
Stylohyoid	Base of styloid process	Body of hyoid (splits round digastric tendon)	VII	Pulls hyoid up and back
Digastric, anterior belly	Digastric fossa of mandible	Intermediate tendon to hyoid	Nerve to mylohyoid (V3)	Opens mouth; raises hyoid
Digastric, posterior belly	Mastoid notch	Intermediate tendon to hyoid	VII	Raises hyoid up and back
Mylohyoid	Mylohyoid line of mandible	Midline raphe and body of hyoid	Nerve to mylohyoid (V3)	Supports and raises floor of mouth
Geniohyoid	Inferior mental spines	Body of hyoid	C1 via XII	Pulls hyoid forward; depresses mandible
Sternohyoid	Manubrium and sternoclavicular joint	Body of hyoid	Ansa cervicalis (C1–C3)	Depresses hyoid
Omohyoid	Upper border of scapula near suprascapular notch	Lower border of hyoid body	Ansa cervicalis (C1–C3)	Depresses and fixes hyoid
Thyrohyoid	Oblique line of thyroid cartilage	Greater horn and body of hyoid	C1 via XII	Depresses hyoid or raises larynx
Sternothyroid	Back of manubrium	Oblique line of thyroid cartilage	Ansa cervicalis (C1–C3)	Pulls larynx down

Gray's Atlas. Infrahyoid (strap) muscles are retracted or divided in thyroidectomy; division high (near the larynx) spares their nerves, which enter low.

M Mnemonics

Tongue muscles

All -glossus muscles are XII except Palatoglossus

Palatoglossus is supplied by the vagus

Taste nerves

7, 9, 10 front to back

Chorda tympani (anterior 2/3), IX (posterior 1/3), X (epiglottis)

Soft palate muscles

Only the Tensor is Trigeminal

Tensor veli palatini (V3); levator, palatoglossus, palatopharyngeus and musculus uvulae are vagus

Q Rapid recall

1. **Boundaries of the oral cavity?** From the lips to the junction of hard and soft palate above and the circumvallate papillae below.
2. **Which extrinsic tongue muscle is not supplied by XII?** Palatoglossus (vagus via the pharyngeal plexus).
3. **Sensation and taste of the anterior two-thirds of the tongue?** Sensation: lingual nerve (V3). Taste: chorda tympani (VII).
4. **Lymphatic drainage of the tongue tip?** Submental nodes (level IA).
5. **Which soft palate muscle is supplied by V3?** Tensor veli palatini.
6. **Attachments of the pterygomandibular raphe?** Pterygoid hamulus to the mylohyoid line; buccinator in front, superior constrictor behind.

7. **Cancer subsites of the oral cavity?** Lip, buccal mucosa, upper and lower alveolus, retromolar trigone, floor of mouth, oral tongue, hard palate.
8. **Lymph drainage of the central lower lip?** Submental nodes (IA), often bilateral.
9. **Where does the floor of mouth drain?** Anterior: IA/IB, crossing the midline. Posterior: levels II–III.
10. **What is the retromolar trigone?** Mucosa over the anterior ramus behind the last molar.
11. **Commonest site of minor salivary gland tumours?** Hard palate.
12. **Number of circumvallate papillae?** About 8–12, in front of the sulcus terminalis.
13. **Which papillae have no taste buds?** Filiform papillae.
14. **Main protruder of the tongue?** Genioglossus.
15. **What marks the boundary of oral and pharyngeal tongue?** The sulcus terminalis with the circumvallate papillae.
16. **Which nerve supplies taste to the palate?** Greater petrosal nerve (VII), via palatine nerves.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.